Answer Key: Practice Exam Version 3

Multiple Choice

#	Ans	Concept tested	Why
1	**A**	DNA content through meiosis	Homologs separated (2N to 1N) but sister chromatids are still joined, so DNA is 2X.
2	**D**	Reading ploidy and replication	Two of each size = homologous pairs = diploid. Single rods (not X-shaped) = unreplicated.
3	**C**	Mitosis vs meiosis	Both separate sister chromatids. The other choices happen only in meiosis I.
4	**E**	Independent assortment vs segregation	Random orientation of each pair at metaphase I. (Anaphase I is segregation.)
5	**B**	Incomplete dominance ratios	Rr x Rr gives 1 RR : 2 Rr : 1 rr, and each genotype looks different, so 1:2:1.
6	**A**	Gametes carry one allele per gene	A gamete needs exactly one allele of each of the three genes.
7	**C**	X-linked crosses	Sons get their X from Mom: half get the normal X, half the hemophilia X.
8	**D**	Pedigree analysis	Affected x affected giving an unaffected child means both parents carry a hidden normal allele, so the trait is dominant.
9	**B**	Linkage	Linked genes travel together on one chromosome instead of sorting independently.
10	**E**	Modified ratios = epistasis	Adds to 16 (two genes) but isn't 9:3:3:1, so one gene is masking the other.
11	**A**	Gene expression vs phenotypic expression	Recessive alleles are usually still expressed; the dominant protein masks their effect.
12	**C**	Multiple alleles	One gene with three alleles (I^A, I^B, i). Height = polygenic, PKU = pleiotropy, 9:3:4 = epistasis.
13	**D**	Chargaff's rules	A = T = 30%, so G + C = 40% and C = 20%.
14	**B**	Meselson-Stahl predictions	After round 2: half intermediate (one old heavy strand each), half fully light.
15	**C**	Leading vs lagging strand	Polymerase only adds to a 3' end, and the strands are antiparallel.
16	**A**	Central Dogma	Transcription = DNA to RNA. Translation = RNA to protein. Replication = DNA to DNA.
17	**E**	Predicting lac operon mutations	Nothing can block transcription, and with no glucose CAP-cAMP boosts it, so it's on even without lactose.
18	**D**	Template strand to mRNA	Complement of the template, antiparallel, with U instead of T.
19	**B**	Prokaryotic vs eukaryotic genomes	Compact, nearly all functional, and size tracks metabolic capability.
20	**C**	Human genome composition	Under 2% codes for protein. ~26% introns, ~50% transposable elements.
21	**A**	Reading the genetic code	Start at the first AUG: AUG GCU UGG UAG = Met-Ala-Trp-stop.

#	Ans	Concept tested	Why
22	**D**	Mutation types	An early frameshift changes every codon after it.
23	**B**	Prokaryotic vs eukaryotic expression	No nucleus, so ribosomes translate mRNA while it's still being transcribed.
24	**C**	Promoter vs coding mutations / exome	Promoter mutations reduce the amount of a normal protein and aren't in the exome.

Answer spread: A = 5, B = 5, C = 6, D = 5, E = 3

Short Answer (model answers)

25. Homozygous: Having two identical alleles of a gene at a locus (e.g., AA or aa).

26. Carrier: A heterozygous individual who has a normal phenotype but carries one copy of a recessive disease allele and can pass it to offspring.

27. Promoter: A DNA sequence near the start of a gene where transcription factors and RNA polymerase bind to begin transcription.

28. Okazaki fragments: Short DNA fragments made on the lagging strand, each starting with an RNA primer, because DNA polymerase only builds 5' to 3'; they are later joined by DNA ligase.

29. Nondisjunction: The failure of homologous chromosomes (meiosis I) or sister chromatids (meiosis II/mitosis) to separate, producing cells with extra or missing chromosomes (aneuploidy).

30. Pleiotropy: When a single gene affects multiple, seemingly unrelated traits (e.g., PKU).

Essay: what a full-credit answer includes

1. Genes and operon: lacZ = beta-galactosidase (cuts lactose into glucose + galactose); **lacY** = permease (brings lactose into the cell); **lacA** = transacetylase. An operon is related genes transcribed from **one promoter into one mRNA**, so they are **co-regulated** (turned on/off together), which saves energy.

2. Regulation: The **repressor** (made by lacI, constitutive) binds the **operator** and blocks RNA polymerase (**negative** regulation). Lactose (the inducer) binds the repressor **allosterically** so it releases the operator. Low glucose raises **cAMP**; **CAP-cAMP** binds near the promoter and recruits RNA polymerase (**positive** regulation).

- Glucose +, lactose -: **off** (repressor bound)
- Glucose +, lactose +: **low** (repressor off, but no CAP boost)
- Glucose -, lactose -: **off** (repressor still blocks, even though CAP is bound)
- Glucose -, lactose +: **high** (repressor off AND CAP-cAMP bound)

3. Mutations: (a) **No lacI**: no repressor, so the operon is transcribed even without lactose (constitutive; high when glucose is absent, low when present). (b) **Repressor can't bind lactose**: it never leaves the operator, so the operon is **always off**. (c) **No CAP**: no recruitment of RNA polymerase, so at most **low** transcription even when glucose is absent and lactose is present.

4. Eukaryotes: Co-regulated genes are scattered (even on different chromosomes) but share the **same regulatory element**, so one transcription factor turns them all on (e.g., **heat shock genes** and the heat shock factor, or estrogen-responsive genes and the estrogen receptor). Eukaryote-only layers: **chromatin** (HATs open it, HDACs close it), **alternative splicing**, mRNA processing/export/stability, and distant **enhancers/silencers**.